



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)

Assignment-3
CSE 3812: Artificial Intelligence Lab
Total Marks: **10**

Assignment Title: Predicting Student Exam Success Using KNN

Assignment Instructions:

- Submit 2 files: 1 file containing the code for KNN and 1 pdf containing the rest of the requirements mentioned below.
- Don't submit a zip file.
- Rename the files as ***Student_Id_KNN.ipynb***

Scenario Overview

Imagine you are working as a data analyst at a tutoring center. You have student data which includes each student's study hours, sleep hours, and prior average grades. Based on these features, you want to predict whether a student will **pass** or **fail** an upcoming exam. You decide to use the **k-Nearest Neighbors (KNN)** algorithm for this classification task.

Assignment Tasks

1. Data Collection/Creation

You have been given a small sample dataset (see **Sample Dataset** section below). In practice, you might have a larger dataset with more detailed features, but for this assignment, we will use a simplified version.

2. Data Exploration

- Print the first few rows of the dataset.
- Briefly describe the variables:
 - **Hours_Studied**: The number of hours the student spends studying per day in the week leading up to the exam.

- **Hours_Slept:** The number of hours of sleep the student gets per day in the week leading up to the exam.
 - **Prior_Grade:** The average of the student's grades before this exam (e.g., on a scale of 0 to 100).
 - **Result:** The final label indicating whether the student **passed** or **failed** the exam.
3. **Preprocessing**
 - Identify any missing values (if any). In practice, you might drop them or fill them with a suitable strategy (like mean or median).
 - Consider whether you need to scale the data, since KNN performance can be sensitive to different feature scales.
 4. **Train/Test Split**
 - Split the dataset into a **training set** and a **test set**. A common practice is an 80%/20% or 70%/30% split.
 5. **Model Training**
 - Use **k-Nearest Neighbors** from `sklearn.neighbors` (e.g., `KNeighborsClassifier`).
 - Choose a value of **k** (e.g., $k=3$ or $k=5$). Explain how you decided on that value.
 6. **Evaluation**
 - Predict the results on the **test set**.
 - Calculate at least two evaluation metrics (e.g., **accuracy**, **precision**, **recall**, **F1-score**).
 - Create a **confusion matrix** to see how many students were correctly classified as pass or fail.
 7. **Hyperparameter Tuning (Optional)**
 - Experiment with different values of **k** to see how it affects performance.
 - (Optional) Experiment with distance metrics (e.g., Euclidean vs. Manhattan).
 8. **Conclusion**
 - Summarize your findings:
 - Which value of **k** gave the best results?
 - What is the overall accuracy?
 - What real-life factors might affect this model's performance?

Sample Dataset

Below is a small example dataset. You can copy this into a CSV file (e.g., `student_data.csv`) or directly load it as a pandas DataFrame.

Hours_Stu died	Hours_Sl ept	Prior_Gr ade	Resu lt
2.0	7.0	70	fail
3.5	6.0	65	fail
5.0	8.0	80	pass
1.0	5.0	50	fail
4.0	6.5	78	pass
6.0	8.5	85	pass
1.5	6.0	60	fail
3.0	7.0	75	pass
2.5	5.0	55	fail
4.5	7.5	82	pass

Interpretation:

- A student who studies 2.0 hours/day, sleeps 7.0 hours/day, and had a prior grade of 70 ended up failing this exam.
- A student who studies 5.0 hours/day, sleeps 8.0 hours/day, and had a prior grade of 80 passed.

Example Inputs and Outputs

Input

Suppose you have trained your KNN model with **k=3** using the data above. Now you want to predict for a **new student** who:

- Studies **3 hours/day**,
- Sleeps **6.5 hours/day**,
- Has a prior grade of **77**.

You run `knn.predict([[3, 6.5, 77]])`.

Output

Depending on how the model is trained, you might get:

`['pass']`

or

`['fail']`